COVER: Knowledge-Grounded Coverage and Conflict Reasoning for Natural Language Inference in English and Portuguese

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Abstract

[TODO: Write last.]

1 Introduction

[TODO: Motivation; gap; contributions.]

2 Related Work

2.1 External knowledge in NLI

Lexical knowledge has long been used to decide whether a hypothesis word is licensed by the premise. KIM [3] feeds WordNet relations (synonymy, antonymy, hypernymy, hyponymy, co-hyponymy) as pairwise features into the co-attention, local inference and composition steps of ESIM, and Glockner et al. [12] showed that models trained on SNLI fail on pairs that differ by a single lexical substitution, precisely the cases where such knowledge should help.

With pretrained Transformers, knowledge is usually injected either as text or as entity embeddings: K-BERT [18] appends knowledge-graph triples to the input and restricts their visibility in self-attention, while KnowBert [29] and ERNIE [48] link mentions to knowledge-base entity embeddings.

Graph-based approaches retrieve a subgraph for the pair and encode it with a graph neural network whose output is fused with the sentence representation [17]. [TODO: add the references of the KGAnet and KGNLI baselines.]

A common limitation is that the retrieved evidence is used as an additional feature vector: it does not change *how* the decision is reached, and its effect on strong pretrained encoders is often small. In COVER, typed relations bias the premise–hypothesis alignment and enter the messages that each hypothesis word receives, and the resulting word-level evidence is combined by explicit logical rules.

2.2 Structure inside pretrained encoders

A second line of work integrates graphs with pretrained language models more tightly. QA-GNN [45] reasons jointly over a question–answer context node and a knowledge-graph subgraph, GreaseLM [47] exchanges information between the language model and a graph neural network at several layers, and structural adapters [32] insert graph convolutions between frozen Transformer blocks.

These methods target question answering or generation over knowledge or semantic graphs. COVER instead builds its graph over the contextual states of the encoder itself, connecting premise and hypothesis words through lexical relations, and keeps the encoder untouched: the graph operates as a separate expert whose output is added to the classifier logits.

2.3 Alignment and natural logic

Alignment between premise and hypothesis is a classic ingredient of NLI, from phrase-based aligners [22] to the soft attention of the decomposable attention model [28] and ESIM [2], whose matching features ($[\boldsymbol{x}; \boldsymbol{m}; \boldsymbol{x} \odot \boldsymbol{m}; |\boldsymbol{x} - \boldsymbol{m}|]$) we reuse in the update of Equation (6).

Natural logic [21] goes further and composes lexical relations along an alignment, taking into account the monotonicity of the context; NaturalLI [1] uses it for large-scale inference, and recent work makes the composition differentiable [8] or trains it with reinforcement learning and lexical resources [9].

These systems depend on external aligners or polarity parsers that are mostly available for English. COVER uses a learned soft alignment with a *null* target and does not model monotonicity explicitly; the weakness this causes on downward-monotone contexts is discussed in Section 4.

2.4 Atomic and probabilistic inference

The aggregation in COVER is closest to models that decide NLI from the parts of the hypothesis. Glickman et al. [11] defined the probability that a text entails a hypothesis as the product, over hypothesis words, of the probability that each word is entailed by some text word, an idea later extended by Shnarch et al. [35].

In neural NLI, SLR-NLI [37] treats hypothesis spans as logical atoms, predicts a label for each span from sentence-level supervision, and derives the pair label with rules such as "a contradiction requires at least one contradicting span"; it retains in-distribution accuracy and improves robustness when training data is scarce.

EPR [43] aligns phrases of the two sentences, predicts a label per aligned pair and induces the sentence label with differentiable fuzzy-logic formulas, and FGLR [38] applies the same principle to facts generated by a large language model.

COVER shares the decision rule of these models (any conflict yields contradiction; full support yields entailment) but differs in three respects. First, its atoms are contextual token nodes rather than spans, phrases or generated facts, so no chunker or generator is needed. Second, the alignment

and the messages of these atoms are shaped by typed lexical relations.

Third, the structured distribution is fused with, rather than substituted for, the sentence-level classifier, which keeps phenomena that the rules cannot express within reach of the encoder. [TODO: confirm in the SLR-NLI and EPR papers whether either combines its prediction with a sentence-level classifier or uses external knowledge.]

2.5 Robustness and diagnostic evaluation

NLI models exploit annotation artifacts [15] and shallow heuristics such as lexical overlap [24], so in-distribution accuracy overstates their inferential ability.

Diagnostic sets address this by annotating pairs with the phenomenon they test: the GLUE/SuperGLUE diagnostic [40, 41], which we use as AX-b, covers lexical semantics, predicate–argument structure, logic and world knowledge, and MED [44] shows that models fine-tuned on standard data handle upward but not downward monotonicity. Following this line, we report results on out-of-distribution and diagnostic test sets in addition to in-distribution accuracy, and analyse AX-b by phenomenon.

2.6 NLI for Portuguese

Portuguese NLI has been driven by shared tasks and translated resources: ASSIN [10] annotates entailment and paraphrase in news, SICK-BR [30] translates SICK, and ASSIN 2 [31] builds on SICK-BR with a binary entailment label. InferBR [39] provides three-way labels and is the main benchmark of this work.

Most recent systems fine-tune Portuguese or multilingual encoders such as BERTimbau [36], Albertina [33] and XLM-R [5], and cross-lingual evaluation typically relies on XNLI [4], which does not include Portuguese.

Lexical-semantic resources for Portuguese are comparatively rich, from Onto.PT [14] to PT-LKB [13], which merges ten resources, but to our knowledge they have not been integrated into the inference mechanism of Transformer-based NLI models. [TODO: check this claim with a targeted search before submission.] COVER uses PT-LKB in this way, and we also test whether the effect transfers between English and Portuguese.

3 Methodology

3.1 Task formulation

Given a premise $\mathcal{P}$ and a hypothesis $\mathcal{H}$, natural language inference (NLI) asks whether $\mathcal{P}$ entails $\mathcal{H}$ (E), contradicts it (C), or neither (N). Some corpora only distinguish entailment from its absence; we keep a single three-way output space $\mathcal{Y} = \{\text{E}, \text{N}, \text{C}\}$ and evaluate two-way corpora through the induced binary decision E *vs.* ¬E (Section 3.6).

Our design starts from a simple observation about the label semantics. A hypothesis is entailed when *every* part of it is supported by the premise and *no* part of it conflicts with the premise; it is contradicted as soon as *one* part conflicts; otherwise it is neutral.

This view goes back to probabilistic lexical entailment [11], where $P(\mathcal{H} \mid \mathcal{P})$ is a product over hypothesis words, and to recent atomic-inference models that predict a label per hypothesis span and combine them with logical rules [37, 43]. COVER implements it over *contextual token states* and grounds the per-word decisions in *typed lexical knowledge*.

3.2 The COVER architecture

Figure 1 gives an overview. COVER keeps the standard cross-encoder classifier and adds a structured expert whose output is a log-probability distribution over $\mathcal{Y}$. The expert has three parts: a token graph built from symbolic resources (Section 3.2.2), bipartite relational attention over contextual states (Section 3.2.3), and a coverage–conflict head with a logical aggregation (Section 3.2.4). The two are fused in logit space (Section 3.2.6).

3.2.1 Encoder

The pair is encoded jointly as `[CLS]` $\mathcal{P}$ `[SEP]` $\mathcal{H}$ `[SEP]` by a pretrained Transformer, yielding contextual states $\boldsymbol{h}_t \in \mathbb{R}^d$ for every subword t and a pooled state $\boldsymbol{h}_{[\text{CLS}]}$ (the pooler output when the encoder has one, otherwise the first token). The baseline classifier is

$$\boldsymbol{\ell}_{[\text{CLS}]} = \boldsymbol{W}_c \, \boldsymbol{h}_{[\text{CLS}]} + \boldsymbol{b}_c \in \mathbb{R}^3. \tag{1}$$

COVER is encoder-agnostic; we use RoBERTa [19] for English, BERTimbau [36] for Portuguese and XLM-R [5] for the cross-lingual setting (Section 3.6).

3.2.2 Token graph

The graph is aligned with the encoder's subword sequence, so that symbolic evidence attaches to contextual states rather than to static word embeddings. A word becomes a node if it is a content word (not punctuation and not in a language-specific stopword list) or a logical operator; the node is represented by the state of its *first* subword. We write $\mathcal{P}$ and $\mathcal{H}$ also for the sets of premise and hypothesis nodes.

Each word also receives an operator type $o_i \in \mathcal{O}$ from language-specific lexicons for English and Portuguese, with 13 types: *none*, negation, universal, existential, downward-monotone, restrictive, conditional, disjunction, conjunction, comparative, modal, temporal and number (e.g. *not, no, without* / *não, nenhum, sem* are negations).

Portuguese *se*/*caso* is tagged as conditional only at the start of a clause, since it is otherwise a reflexive pronoun or a noun.

Premise and hypothesis are connected by typed lexical relations. For every premise–hypothesis word pair, a relation $r_{ij} \in \mathcal{R}$ is assigned by the first rule that fires: (i) *number-equal* or *number-conflict* when both words are numerals; (ii) *identical* surface form; (iii) *antonym*, *hypernym*, *hyponym* or *synonym* from a lexical resource, looked up over a small set of lemma candidates of each word; (iv) *same-stem* under a light suffix stripper; otherwise *none*.

Hypernymy is directional: $r_{ij} = \textit{hypernym}$ means the hypothesis word j is more general than the premise word

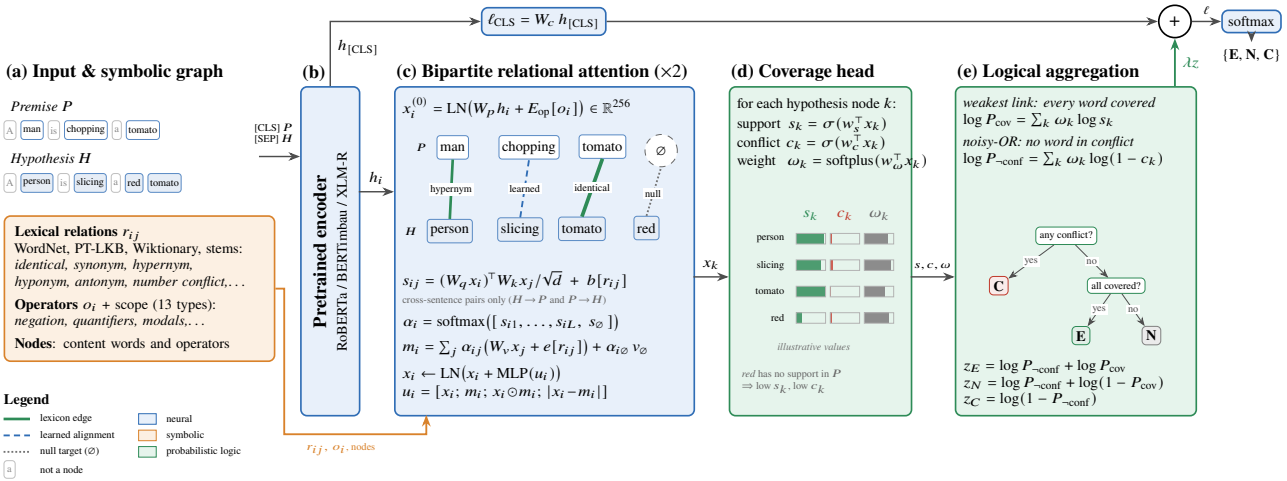


Figure 1: COVER architecture. (a) Content words and logical operators of $\mathcal{P}$ and $\mathcal{H}$ become graph nodes; typed lexical relations r_{ij} link premise and hypothesis nodes. (b) A pretrained encoder reads the pair jointly. (c) Two bipartite relational attention layers refine each node with its evidence on the other side, including a learned *null* target. (d) Every hypothesis node k receives a support s_k, a conflict c_k and an importance ω_k. (e) A weakest-link coverage and a noisy-OR conflict define a structured distribution z, added to the [CLS] logits. [TODO: update panel (a): operator scope edges are built but not used by COVER.]

i ($i \sqsubset j$), and the reverse edge r_{ji} is the inverse relation. Stopwords never receive lexical relations.

The resources are WordNet 3.0 [25] and Wiktionary [46] for English, and PT-LKB [13, 27], which merges ten Portuguese lexical resources, and Wiktionary for Portuguese. Pairs listed with both hypernym directions in the merged resources are discarded instead of guessed, since a wrong direction swaps forward and reverse entailment. From WordNet we keep single-word lemmas and the three most frequent synsets per part of speech, to limit edges from rare senses.

3.2.3 Bipartite relational attention

Node states are projected to a smaller space and typed by their operator,

$$\boldsymbol{x}_i^{(0)} = \mathrm{LN}\big(\boldsymbol{W}_p\, \boldsymbol{h}_i + \boldsymbol{E}_{\mathrm{op}}[o_i]\big) \in \mathbb{R}^{d_g}, \tag{2}$$

with $d_g = 256$. Each of $L = 2$ layers lets every premise node attend to the hypothesis nodes and every hypothesis node attend to the premise nodes (cross-sentence pairs only), plus a learned *null* target $\varnothing$ for words without a counterpart. For a node i and a node j on the other side,

$$e_{ij} = \frac{(\boldsymbol{W}_q \boldsymbol{x}_i)^\top \boldsymbol{W}_k \boldsymbol{x}_j}{\sqrt{d_g}} + b[r_{ij}], \tag{3}$$

$$\boldsymbol{\alpha}_i = \mathrm{softmax}\left([\, e_{i1}, \ldots, e_{iN}, e_{i\varnothing} \,]\right), \tag{4}$$

$$\boldsymbol{m}_i = \sum_j \alpha_{ij}\big(\boldsymbol{W}_v \boldsymbol{x}_j + \boldsymbol{e}[r_{ij}]\big) + \alpha_{i\varnothing}\, \boldsymbol{v}_\varnothing, \tag{5}$$

$$\boldsymbol{x}_i \leftarrow \mathrm{LN}\big(\boldsymbol{x}_i + \mathrm{MLP}([\boldsymbol{x}_i; \boldsymbol{m}_i; \boldsymbol{x}_i \odot \boldsymbol{m}_i; |\boldsymbol{x}_i - \boldsymbol{m}_i|])\big). \tag{6}$$

The relation enters twice: as a scalar bias $b[r]$ on the alignment score, which steers *where* a word looks, and as a vector $\boldsymbol{e}[r]$ added to the message, which tells the receiving word *what kind* of evidence it found (e.g. an antonym rather than a synonym). The bias is initialised to 2 for identical, synonym, hypernym, hyponym, antonym and number relations, 1 for same-stem and 0 otherwise, so that lexicon edges are preferred alignments at the start of training; all relation parameters are learned.

The update in Equation (6) follows the matching features of ESIM [2]; $e_\varnothing$ and $\boldsymbol{v}_\varnothing$ are learned, and layers do not share weights. Unlike pairwise lexical features [3], the relations here act on contextual states and are refined over two rounds of alignment.

3.2.4 Coverage and conflict head

After the last layer, each hypothesis node $k \in \mathcal{H}$ is scored independently:

$$s_k = \sigma(\boldsymbol{w}_s^\top \boldsymbol{x}_k + b_s), \qquad \text{(support)} \tag{7}$$

$$c_k = \sigma(\boldsymbol{w}_c^\top \boldsymbol{x}_k + b_c), \qquad \text{(conflict)} \tag{8}$$

$$\omega_k = \mathrm{softplus}(\boldsymbol{w}_\omega^\top \boldsymbol{x}_k + b_\omega). \qquad \text{(importance)} \tag{9}$$

s_k is the probability that word k is supported by the premise, c_k the probability that it conflicts with the premise, and $\omega_k \geq 0$ how much the word matters for the decision (function-like content words can be down-weighted). Biases are initialised to $b_s = 2$ and $b_c = -3$, i.e. words start as supported and non-conflicting.

3.2.5 Logical aggregation

The word-level scores are combined by two logical rules. The hypothesis is covered if *all* of its words are supported (weakest link) and is free of conflict if *none* of its words conflicts (noisy-OR):

$$\log P_{\mathrm{cov}} = \textstyle\sum_{k \in \mathcal{H}} \omega_k \log s_k, \tag{10}$$

$$\log P_{\neg\mathrm{conf}} = \textstyle\sum_{k \in \mathcal{H}} \omega_k \log(1 - c_k). \tag{11}$$

With $\omega_k = 1$, Equation (10) is the product of word-level support probabilities of Glickman et al. [11], and Equation (11) is a noisy-OR over word-level conflicts. The three

labels follow the decision tree "any conflict ⇒ C; otherwise all covered ⇒ E; otherwise N":

$$\begin{aligned} z_{\text{E}} &= \log P_{\neg\text{conf}} + \log P_{\text{cov}}, \\ z_{\text{N}} &= \log P_{\neg\text{conf}} + \log(1 - P_{\text{cov}}), \\ z_{\text{C}} &= \log(1 - P_{\neg\text{conf}}). \end{aligned} \tag{12}$$

Since $P_{\neg\text{conf}} P_{\text{cov}} + P_{\neg\text{conf}}(1 - P_{\text{cov}}) + (1 - P_{\neg\text{conf}}) = 1$, z is a normalised log-distribution by construction; no extra softmax or threshold is needed.

Atomic-inference models apply similar rules with discrete span labels [37] or fuzzy logic over aligned phrases [43]; COVER differs in that the atoms are contextual token nodes whose alignment is shaped by typed lexical knowledge (Equations (3) and (5)), and in that the result is fused with, rather than substituted for, the sentence-level classifier. The per-word values (s_k, c_k, ω_k) also serve as an explanation of each prediction.

3.2.6 Fusion with the sentence classifier

The structured distribution is added to the [CLS] logits with a learned scale λ (initialised to 1):

$$\boldsymbol{\ell} = \boldsymbol{\ell}_{[\text{CLS}]} + \lambda\, z, \qquad \hat{y} = \arg\max_{y}\ \ell_y. \tag{13}$$

This is a product of experts: the coverage–conflict expert can overrule the encoder when its evidence is strong (e.g. a confident conflict on one word), and defers to it otherwise, so that phenomena the head cannot express are still handled by the encoder.

3.3 Training objective

We use a set-valued negative log-likelihood $\mathcal{L}_{\text{set}}(\boldsymbol{u}; Y) = -\log \sum_{y \in Y} \text{softmax}(\boldsymbol{u})_y$ over a set Y of admissible labels, applied to the fused logits and, as an auxiliary term, to the structured expert alone:

$$\mathcal{L} = \mathcal{L}_{\text{set}}(\boldsymbol{\ell}; Y) + \beta\, \mathcal{L}_{\text{set}}(z; Y), \qquad \beta = 0.15. \tag{14}$$

The auxiliary term keeps z a meaningful classifier on its own instead of a residual correction. With a single gold label, $Y = \{y^*\}$ and Equation (14) is the usual cross-entropy, which is the case in all experiments reported here.

The set formulation also allows training on corpora whose negative class merges neutral and contradiction (e.g. the NONE label of ASSIN), with $Y = \{\text{N}, \text{C}\}$; we leave this mixed-corpus training to future work.[TODO: keep or drop depending on the final experiments.]

The encoder and the new parameters are optimised with AdamW [20] and a linear warm-up/decay schedule; the graph and head parameters use a learning rate 10× larger than the encoder's, since they start from scratch.

3.4 Ablations

To identify which components produce the effect, we compare COVER with five variants that remove one component each while keeping everything else, including the auxiliary loss, identical (Figure 2):

w/o lexicon all relations and operators are set to *none* ($r_{ij} = \textit{none}$, $o_i = \textit{none}$): the graph and head remain, but alignment is learned from the encoder states only.

w/o graph $L = 0$: the head reads $\boldsymbol{x}^{(0)}$ directly; this also removes the relation-typed attention, hence the lexicon edges.

w/o conflict coverage only: $z_{\text{E}} = \log P_{\text{cov}}$ and $z_{\text{N}} = z_{\text{C}} = \log(1 - P_{\text{cov}}) + \log \frac{1}{2}$.

w/o coverage conflict only: $z_{\text{C}} = \log(1 - P_{\neg\text{conf}})$ and $z_{\text{E}} = z_{\text{N}} = \log P_{\neg\text{conf}} + \log \frac{1}{2}$.

MLP head the logical aggregation (Section 3.2.5) is replaced by an MLP over the mean and max of the hypothesis node states, followed by a log-softmax; this controls for the capacity of the head.

The [CLS] baseline (Equation (1) alone) is trained with the same hyperparameters.

3.5 Knowledge-graph baselines

We also compare with three knowledge-enhanced NLI architectures, re-implemented on the same encoders and trained under the same protocol and seeds. K-BERT-style injection [18] appends the retrieved relations to the premise as text, with a 160-token window so that they are not truncated. [TODO: add KGAnet and KGNLI references.]

The KGAnet-style model encodes a relation-labelled word graph with a relation-aware graph attention network, and the KGNLI-style model keeps separate semantic and knowledge branches. All three take their knowledge from rule-based relations (numbers, operators, scope) and English WordNet, so on Portuguese text their lexical evidence reduces to word identity.

3.6 Experimental protocol

3.6.1 Settings and data

We evaluate three settings (Figure 3 and Table 1). *English*: RoBERTa-large [19] trained on SICK [23], tested on the SICK test set, on the SuperGLUE diagnostic AX-b [41], which annotates pairs with linguistic phenomena (lexical semantics, predicate–argument structure, logic, knowledge), and on three challenge sets described below.

Portuguese: BERTimbau-base [36] trained on InferBR [39] [TODO: check InferBR citation], tested on the InferBR test set and, out of distribution, on the ASSIN [10], ASSIN 2 [31] and SICK-BR [30] test sets.

Cross-lingual: XLM-R-base [5] trained on SICK or on InferBR and tested on all nine test sets, with the union of the English and Portuguese lexicons in both directions. Relation types are shared across languages, so relation parameters learned in one language are applied to edges of the other.

The English challenge sets each isolate one kind of inference. Breaking NLI [12] changes a single word of an SNLI premise (antonyms, co-hyponyms, hypernyms), a direct test of lexical knowledge and of the conflict detector.

ConjNLI [34] tests conjunctive sentences; we use its labelled development set, since the test labels are not public.

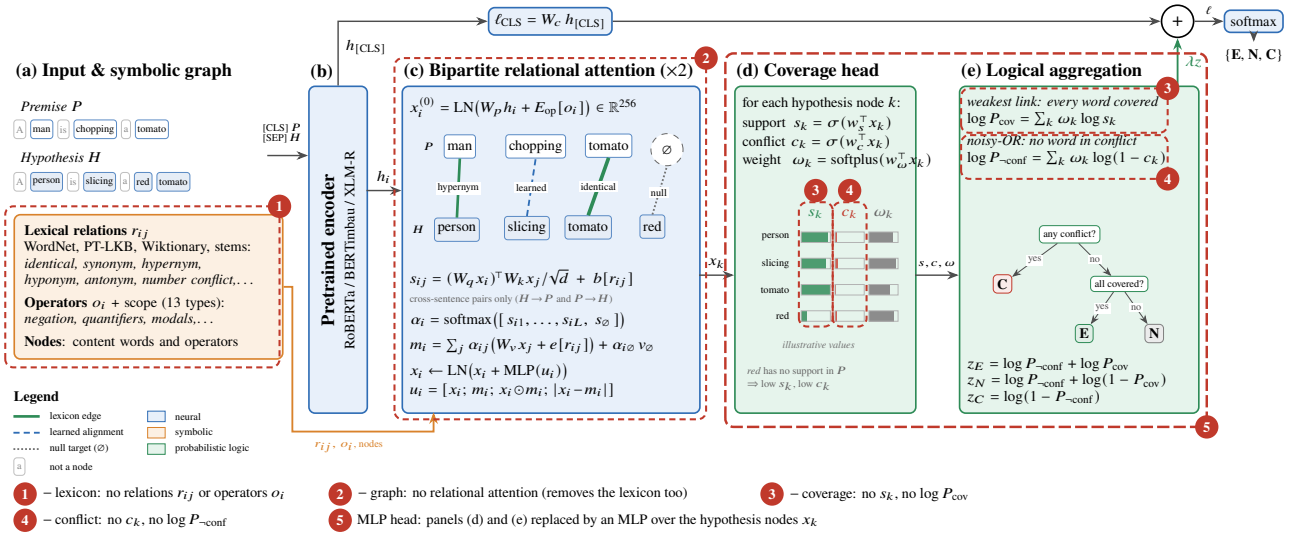


Figure 2: What each ablation removes (red dashed boxes), drawn on the architecture of Figure 1; the numbers refer to the key below the diagram. The [CLS] baseline keeps only the encoder and the top ℓ_{CLS} path.

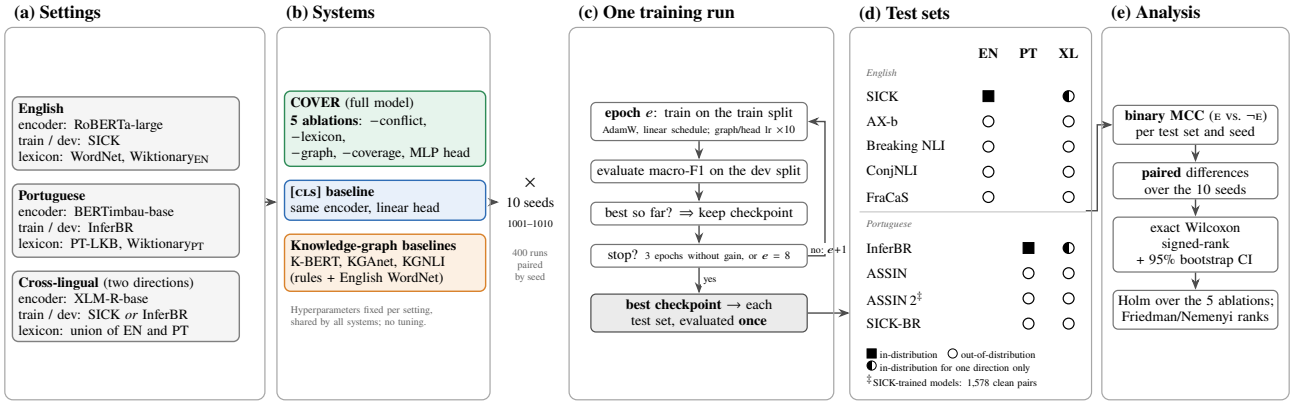


Figure 3: Experimental design. (a) Three settings, each with its own encoder, training data and lexicon; the cross-lingual setting is run in both directions. (b) All systems of a setting share the same fixed hyperparameters. (c) Each system is trained with 10 seeds; a run keeps the checkpoint with the best development macro-F1 and stops after 3 epochs without improvement (at most 8). (d) The selected checkpoint is evaluated once on every test set of its setting. (e) Systems are compared with paired tests over the seeds.

The FraCaS suite [6] covers formal semantic phenomena such as quantifiers, plurals and attitudes; multiple premises are concatenated, and the 11 problems without a definite answer are discarded.

SICK-BR is the Portuguese translation of SICK; its test partition is aligned pair by pair with the SICK test set, so the same pairs can be compared in both languages. ASSIN 2 was built from SICK-BR, and about 36% of its test pairs (870 of 2,448) are translations of SICK training or trial pairs. Models trained on SICK are therefore scored on the remaining 1,578 pairs of ASSIN 2.

3.6.2 Training details

Hyperparameters are fixed in advance and shared by all variants of a setting; no per-variant tuning is performed, so that differences reflect architecture rather than search budget. We train for at most 8 epochs with batch size 32, weight decay 3.5×10^{-4}, dropout 0.103, gradient clipping at 1.0, and select the checkpoint with the best development macro-F1, with early stopping after 3 epochs without improvement.

The learning rate is 10^{-5} with 6% warm-up for RoBERTa-large, which is unstable at larger rates on small datasets [26], 2.04×10^{-5} with 2.75% warm-up for BERTimbau, and 2×10^{-5} with 6% warm-up for XLM-R. [TODO: max lengths; gradient accumulation for RoBERTa-large (2×16, same effective batch).]

3.6.3 Evaluation

Our main metric, for every test set, is the Matthews correlation coefficient (MCC) of the binary decision E *vs.* ¬E, obtained by merging N and C predictions. It is the only metric defined for all our test sets: AX-b, ASSIN and ASSIN 2 only distinguish entailment from non-entailment, and MCC is the official AX-b metric. It is also robust to the label imbalance of these sets.

The binary view is also required for Breaking NLI: models trained on SICK label most of its contradictions as neutral, following the annotation convention of SICK, so

Dataset	Lang.	Train	Dev	Test	Labels
SICK	EN	4,439	495	4,906	3-way
AX-b	EN	–	–	1,104	2-way
Breaking NLI	EN	–	–	8,193	3-way
ConjNLI	EN	–	–	623	3-way
FraCaS	EN	–	–	335	3-way
InferBR	PT	8,190	633	1,705	3-way
ASSIN	PT	–	–	4,000	2-way[†]
ASSIN 2	PT	–	–	2,448[‡]	2-way
SICK-BR	PT	–	–	4,906	3-way

Table 1: Datasets. Only SICK and InferBR are used for training; all other sets are used for testing only. [†]Paraphrase pairs are counted as entailment. [‡]1,578 pairs for models trained on SICK (pairs translated from SICK training or trial data removed).

only the entailment decision is comparable across training sets. For the in-distribution test sets (SICK and InferBR) we additionally report three-way macro-F1 in the text.

We further break AX-b results down by the coarse categories of the diagnostic and their fine-grained phenomena. With 18–140 pairs per phenomenon, these breakdowns are exploratory.

Every system is trained with the same 10 seeds (1001–1010), and results are paired by seed. The seed fixes all random number generators at the start of a run; because variants create different numbers of parameters, initialisation and data order are not identical across variants for the same seed, so pairing blocks run-level variance only partially.

We compare two systems with the two-sided exact Wilcoxon signed-rank test on the 10 per-seed differences [42] and report the mean paired difference with a 95% percentile bootstrap interval over seeds. Within the ablation family, p-values are corrected with Holm's method [16]. Rankings of more than two systems use the Friedman test with the Nemenyi post-hoc test [7].

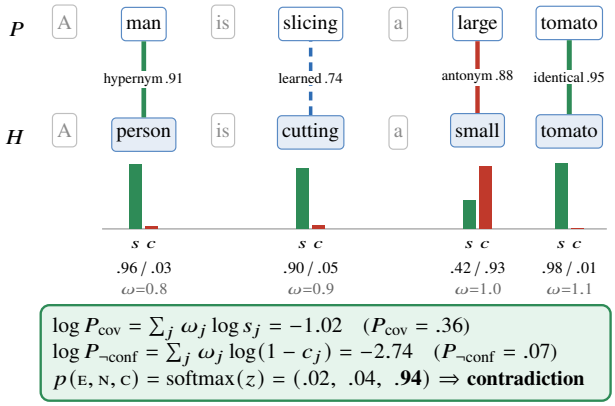


Figure 4: Reading a COVER decision. Each content word of the hypothesis is aligned to a premise word (line width: attention weight; colour and label: lexical relation, dashed when no relation is available). Bars give the support s_j and conflict c_j of each hypothesis word, and ω_j its weight; the box applies the aggregation of Section 3.2. The antonym edge *small–large* produces a high conflict that alone determines the contradiction. [TODO: Illustrative values: replace with node-level scores exported from a trained model.]

4 Results

4.1 Main results

4.2 Ablation

4.3 Error analysis

4.4 Interpretability

(a) English: RoBERTa-large trained on SICK

	Accuracy					MCC				
System	SICK	AX-b	Break.	Conj.	FraCaS	SICK	AX-b	Break.	Conj.	FraCaS
[CLS]	**94.3** ±0.4	69.1 ±2.5	**95.2** ±1.1	63.0 ±6.2	59.6 ±3.3	**86.0** ±0.9	38.2 ±3.7	**80.3** ±3.0	34.2 ±4.5	19.6 ±3.8
K-BERT										
KGAnet										
KGNLI										
COVER	94.2 ±0.2	**71.6**† ±1.5	95.0 ±1.3	**66.7** ±4.7	**65.7**† ±2.6	85.9 ±0.6	**43.1**† ±2.3	79.6 ±4.2	**40.1**† ±3.9	**28.2**† ±3.3

(b) Portuguese: BERTimbau-base trained on InferBR

	Accuracy				MCC			
System	InferBR	ASSIN	ASSIN 2	SICK-BR	InferBR	ASSIN	ASSIN 2	SICK-BR
[CLS]	95.2 ±0.5	82.1 ±2.1	83.9 ±1.6	82.5 ±1.5	89.2 ±1.2	57.2 ±3.4	68.0 ±3.0	58.7 ±3.2
K-BERT								
KGAnet								
KGNLI								
COVER	**95.7** ±0.5	**83.1** ±2.0	**86.3**† ±0.4	**84.6**† ±0.5	**90.4** ±1.0	**60.5**† ±3.7	**72.6**† ±0.9	**64.0**† ±0.8

(c) Cross-lingual EN→PT: XLM-R-base trained on SICK, Portuguese tests

	Accuracy				MCC			
System	InferBR	ASSIN	ASSIN 2	SICK-BR	InferBR	ASSIN	ASSIN 2	SICK-BR
[CLS]	**79.2** ±1.5	**77.3** ±6.7	86.3 ±1.7	**86.5** ±0.6	51.8 ±3.5	**47.4** ±8.7	70.9 ±3.5	**66.4** ±1.3
K-BERT								
KGAnet								
KGNLI								
COVER	78.7 ±1.7	77.0 ±5.2	**86.6** ±0.6	86.0 ±0.5	**52.6** ±2.7	46.1 ±5.4	**71.7** ±1.4	65.8 ±1.2

(d) Cross-lingual PT→EN: XLM-R-base trained on InferBR, English tests

	Accuracy					MCC				
System	SICK	AX-b	Break.	Conj.	FraCaS	SICK	AX-b	Break.	Conj.	FraCaS
[CLS]										
K-BERT										
KGAnet										
KGNLI										
COVER										

Table 2: Main results (mean over 10 seeds, seed standard deviation in small type; best per column in bold). Left block: accuracy; right block: MCC; both for the binary decision entailment *vs.* non-entailment, ×100. Break.: Breaking NLI; Conj.: ConjNLI. (c) and (d) evaluate zero-shot transfer to the other language; the same models on the test sets of their training language are in Table 6. For models trained on SICK, ASSIN 2 excludes pairs translated from SICK training data. †Paired Wilcoxon $p < 0.05$ against [CLS] on the same metric.

(a) English: RoBERTa-large trained on SICK

System	SICK	AX-b	Break.	Conj.	FraCaS
COVER (full)	85.9 ±0.6	43.1 ±2.3	79.6 ±4.2	40.1 ±3.9	28.2 ±3.3
− conflict	+0.2	+2.0	+2.8	+0.4	−0.2
− lexicon	−0.4	+0.2	+0.7	−1.6	+0.5
− graph	+0.2	−1.8	+1.1	−4.8	−0.3
− coverage	+0.2	−6.7*	+0.4	−8.5*	−9.6*
MLP head	−0.2	−5.1*	−3.2	−11.8*	−2.0
[CLS]	+0.1	−4.9†	+0.7	−5.9†	−8.6†

(b) Portuguese: BERTimbau-base trained on InferBR

System	InferBR	ASSIN	ASSIN 2	SICK-BR
COVER (full)	90.4 ±1.0	60.5 ±3.7	72.6 ±0.9	64.0 ±0.8
− conflict	−0.1	+2.4	+1.1*	+1.0*
− lexicon	−0.9	+1.5	−1.5*	−1.3*
− graph	−0.6	−0.3	−2.1*	−1.8*
− coverage	−0.6	−11.9*	−4.0*	−4.2*
MLP head	−0.1	−2.4	−1.4	−0.8
[CLS]	−1.2	−3.3†	−4.6†	−5.3†

(c) Cross-lingual: XLM-R-base

	English tests					Portuguese tests			
System	SICK	AX-b	Break.	Conj.	FraCaS	InferBR	ASSIN	ASSIN 2	SICK-BR
Train SICK (EN)									
COVER (full)	–	–	–	–	–	52.6 ±2.7	46.1 ±5.4	71.7 ±1.4	65.8 ±1.2
− conflict	–	–	–	–	–	−2.4	+0.2	−0.5	+0.3
− lexicon	–	–	–	–	–	−1.9	+1.5	+1.3	+2.1*
− graph	–	–	–	–	–	+0.9	+1.2	+1.0	+2.1*
− coverage	–	–	–	–	–	−1.0	−0.9	−1.4	+0.9
MLP head	–	–	–	–	–	+2.1	+2.1	+0.6	+0.6
[CLS]	–	–	–	–	–	−0.8	+1.3	−0.8	+0.7
Train InferBR (PT)									
COVER (full)						–	–	–	–
− conflict						–	–	–	–
− lexicon						–	–	–	–
− graph						–	–	–	–
− coverage						–	–	–	–
MLP head						–	–	–	–
[CLS]						–	–	–	–

Table 3: Ablation of COVER (binary MCC ×100, same seeds as Table 2). The first row gives the full model (mean, seed standard deviation in small type); the other rows give the mean paired difference to it over seeds, with decreases in red. − conflict: coverage only; − lexicon: no lexical relations or operator lexicon; − graph: no relational attention layers; − coverage: conflict only; MLP head: an MLP over the hypothesis nodes replaces the logical aggregation. In (c) only transfer to the other language is reported (–: test sets in the training language). *Holm-corrected paired Wilcoxon $p < 0.05$ against the full model (over the five ablations); †paired Wilcoxon $p < 0.05$ against the full model.

Phenomenon	n	[CLS]	COVER	Δ
Lexical semantics	368	47.7	50.2	+2.5
Factivity	68	41.4	46.2	+4.8
Lexical entailment	140	39.4	41.9	+2.5
Morphological negation	26	26.7	51.0	+24.3
Named entities	36	69.6	66.7	−2.9
Quantifiers	52	44.4	50.0	+5.6
Redundancy	26	35.2	27.5	−7.7
Symmetry/Collectivity	28	34.6	25.9	−8.7
Predicate–argument structure	424	45.4	51.7	+6.3
Active/Passive	34	46.8	48.4	+1.6
Anaphora/Coreference	58	37.9	50.6	+12.7
Coordination scope	40	41.5	55.1	+13.6
Core args	52	37.4	39.6	+2.2
Datives	20	60.7	62.7	+2.1
Ellipsis/Implicits	34	42.1	54.5	+12.4
Genitives/Partitives	20	57.7	90.7	+33.1
Intersectivity	46	39.5	35.3	−4.2
Nominalization	28	47.9	76.1	+28.2
Prepositional phrases	68	61.7	71.1	+9.3
Relative clauses	32	39.0	43.7	+4.7
Restrictivity	26	12.2	−3.7	−15.8
Logic	364	19.9	26.2	+6.3
Conditionals	32	17.2	20.2	+3.0
Conjunction	40	48.2	65.2	+16.9
Disjunction	38	0.6	−1.1	−1.7
Double negation	28	16.7	39.2	+22.5
Downward monotone	30	−27.6	−33.3	−5.7
Existential	20	37.2	39.5	+2.3
Intervals/Numbers	38	0.9	0.0	−1.0
Negation	82	10.3	11.8	+1.5
Non-monotone	30	10.9	16.3	+5.5
Temporal	32	−3.8	20.3	+24.1
Universal	18	36.3	26.2	−10.1
Upward monotone	34	60.2	55.5	−4.7
Knowledge	284	29.5	31.5	+2.0
Common sense	150	26.1	31.4	+5.3
World knowledge	134	30.2	30.2	+0.0

Table 4: AX-b by phenomenon (binary MCC ×100, English setting, mean over 10 seeds): the four coarse categories of the diagnostic (bold) and their fine-grained phenomena. A pair can carry several phenomena. Δ: COVER minus [CLS], decreases in red. Groups with fewer than 40 pairs are exploratory.

		[CLS]			COVER − [CLS]		
	Gold	E	N	C	E	N	C
SICK (EN)	E	**1265**	137	2	**−3**	+4	−1
	N	141	**2596**	53	−1	**−2**	+3
	C	2	102	**608**	0	+5	**−5**
FraCaS (EN)	E	**120**	66	19	**+27**	−16	−10
	N	45	**41**	12	+7	**−2**	−5
	C	6	10	**17**	0	+4	**−4**
InferBR (PT)	E	**529**	12	27	**+4**	+1	−5
	N	20	**518**	26	0	**−3**	+2
	C	22	32	**519**	−5	+1	**+4**

Table 5: Confusion matrices on the in-distribution test sets and on FraCaS (mean counts per seed; rows: gold label, columns: prediction; diagonal in bold). Left: [CLS]. Right: difference COVER − [CLS] in each cell; red marks where COVER is worse, i.e. fewer correct predictions on the diagonal or more errors off the diagonal.

5 Conclusion

[TODO:]

References

[1] Gabor Angeli and Christopher D. Manning. NaturalLI: Natural logic inference for common sense reasoning. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2014.

[2] Qian Chen, Xiaodan Zhu, Zhen-Hua Ling, Si Wei, Hui Jiang, and Diana Inkpen. Enhanced LSTM for natural language inference. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2017.

[3] Qian Chen, Xiaodan Zhu, Zhen-Hua Ling, Diana Inkpen, and Si Wei. Neural natural language inference models enhanced with external knowledge. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018.

[4] Alexis Conneau, Ruty Rinott, Guillaume Lample, Adina Williams, Samuel R. Bowman, Holger Schwenk, and Veselin Stoyanov. XNLI: Evaluating cross-lingual sentence representations. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2018.

[5] Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020.

[6] Robin Cooper, Dick Crouch, Jan van Eijck, Chris Fox, Josef van Genabith, Jan Jaspars, Hans Kamp, David Milward, Manfred Pinkal, Massimo Poesio, and Steve Pulman. Using the framework. Technical Report LRE 62-051 D-16, The FraCaS Consortium, 1996.

[7] Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.

[8] Yufei Feng, Zi'ou Zheng, Quan Liu, Michael Greenspan, and Xiaodan Zhu. Exploring end-to-end differentiable natural logic modeling. In *Proceedings of the 28th International Conference on Computational Linguistics (COLING)*, 2020.

[9] Yufei Feng, Xiaoyu Yang, Xiaodan Zhu, and Michael Greenspan. Neuro-symbolic natural logic with introspective revision for natural language inference. *Transactions of the Association for Computational Linguistics*, 10, 2022.

[10] Erick R. Fonseca, Leandro Borges dos Santos, Marcelo Criscuolo, and Sandra Maria Aluísio. ASSIN: Avaliação de similaridade semântica e inferência textual. In *Computational Processing of the Portuguese Language (PROPOR), Workshop*, 2016.

[11] Oren Glickman, Ido Dagan, and Moshe Koppel. A probabilistic lexical approach to textual entailment. In *Proceedings of the 19th International Joint Conference on Artificial Intelligence (IJCAI)*, 2005.

[12] Max Glockner, Vered Shwartz, and Yoav Goldberg. Breaking NLI systems with sentences that require simple lexical inferences. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018.

[13] Hugo Gonçalo Oliveira. Learning word embeddings from Portuguese lexical-semantic knowledge bases. In *Computational Processing of the Portuguese Language (PROPOR 2018)*, Lecture Notes in Computer Science, pages 265–271. Springer, 2018. doi: 10.1007/978-3-319-99722-3_27.

[14] Hugo Gonçalo Oliveira and Paulo Gomes. ECO and Onto.PT: a flexible approach for creating a Portuguese wordnet automatically. *Language Resources and Evaluation*, 48(2), 2014.

[15] Suchin Gururangan, Swabha Swayamdipta, Omer Levy, Roy Schwartz, Samuel Bowman, and Noah A. Smith. Annotation artifacts in natural language inference data. In *Proceedings of NAACL-HLT*, 2018.

[16] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6 (2):65–70, 1979.

[17] Pavan Kapanipathi, Veronika Thost, Siva Sankalp Patel, Spencer Whitehead, Ibrahim Abdelaziz, Avinash Balakrishnan, Maria Chang, Kshitij Fadnis, Chulaka Gunasekara, Bassem Makni, Nicholas Mattei, Kartik Talamadupula, and Achille Fokoue. Infusing knowledge into the textual entailment task using graph convolutional networks. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020.

[18] Weijie Liu, Peng Zhou, Zhe Zhao, Zhiruo Wang, Qi Ju, Haotang Deng, and Ping Wang. K-BERT: Enabling language representation with knowledge graph. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2020.

[19] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.

[20] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019.

[21] Bill MacCartney and Christopher D. Manning. An extended model of natural logic. In *Proceedings of the Eighth International Conference on Computational Semantics (IWCS)*, 2009.

[22] Bill MacCartney, Michel Galley, and Christopher D. Manning. A phrase-based alignment model for natural language inference. In *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2008.

[23] Marco Marelli, Stefano Menini, Marco Baroni, Luisa Bentivogli, Raffaella Bernardi, and Roberto Zamparelli. A SICK cure for the evaluation of compositional distributional semantic models. In *Proceedings of the Ninth International Conference on Language Resources and Evaluation (LREC)*, 2014.

[24] Tom McCoy, Ellie Pavlick, and Tal Linzen. Right for the wrong reasons: Diagnosing syntactic heuristics in natural language inference. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019.

[25] George A. Miller. WordNet: A lexical database for English. *Communications of the ACM*, 38(11):39–41, 1995.

[26] Marius Mosbach, Maksym Andriushchenko, and Dietrich Klakow. On the stability of fine-tuning BERT: Misconceptions, explanations, and strong baselines. In *International Conference on Learning Representations (ICLR)*, 2021.

[27] NLP-CISUC. PT-LexicalSemantics: computational resources for Portuguese lexical semantics. `https://github.com/NLP-CISUC/PT-LexicalSemantics`. PT-LKB release used in this work.

[28] Ankur Parikh, Oscar Täckström, Dipanjan Das, and Jakob Uszkoreit. A decomposable attention model for natural language inference. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2016.

[29] Matthew E. Peters, Mark Neumann, Robert Logan, Roy Schwartz, Vidur Joshi, Sameer Singh, and Noah A. Smith. Knowledge enhanced contextual word representations. In *Proceedings of EMNLP-IJCNLP*, 2019.

[30] Livy Real, Ana Rodrigues, Andressa Vieira e Silva, Beatriz Albiero, Bruna Thalenberg, Bruno Guide, Cindy Silva, Guilherme de Oliveira Lima, Igor C. S. Câmara, Miloš Stanojević, Rodrigo Souza, and Valeria de Paiva. SICK-BR: A Portuguese corpus for inference. In *Computational Processing of the Portuguese Language (PROPOR)*, 2018.

[31] Livy Real, Erick Fonseca, and Hugo Gonçalo Oliveira. The ASSIN 2 shared task: A quick overview. In *Computational Processing of the Portuguese Language (PROPOR)*, 2020.

[32] Leonardo F. R. Ribeiro, Yue Zhang, and Iryna Gurevych. Structural adapters in pretrained language models for AMR-to-text generation. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021.

[33] João Rodrigues, Luís Gomes, João Silva, António Branco, Rodrigo Santos, Henrique Lopes Cardoso, and Tomás Osório. Advancing neural encoding of Portuguese with transformer Albertina PT-*. In *Progress in Artificial Intelligence (EPIA)*, 2023.

[34] Swarnadeep Saha, Yixin Nie, and Mohit Bansal. ConjNLI: Natural language inference over conjunctive sentences. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 8240–8252, 2020.

[35] Eyal Shnarch, Jacob Goldberger, and Ido Dagan. A probabilistic modeling framework for lexical entailment. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2011.

[36] Fábio Souza, Rodrigo Nogueira, and Roberto Lotufo. BERTimbau: Pretrained BERT models for Brazilian Portuguese. In *Intelligent Systems: 9th Brazilian Conference (BRACIS)*, 2020.

[37] Joe Stacey, Pasquale Minervini, Haim Dubossarsky, and Marek Rei. Logical reasoning with span-level predictions for interpretable and robust NLI models. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.

[38] Joe Stacey, Pasquale Minervini, Haim Dubossarsky, Oana-Maria Camburu, and Marek Rei. Atomic inference for NLI with generated facts as atoms. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.

[39] TODO. InferBR: a natural language inference dataset in Portuguese, 2024. TODO verify.

[40] Alex Wang, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel Bowman. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In *Proceedings of the 2018 EMNLP Workshop BlackboxNLP*, 2018.

[41] Alex Wang, Yada Pruksachatkun, Nikita Nangia, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. SuperGLUE: A stickier benchmark for general-purpose language understanding systems. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[42] Frank Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.

[43] Zijun Wu, Zi Xuan Zhang, Atharva Naik, Zhijian Mei, Mauajama Firdaus, and Lili Mou. Weakly supervised explainable phrasal reasoning with neural fuzzy logic. In *International Conference on Learning Representations (ICLR)*, 2023.

[44] Hitomi Yanaka, Koji Mineshima, Daisuke Bekki, Kentaro Inui, Satoshi Sekine, Lasha Abzianidze, and Johan Bos. Can neural networks understand monotonicity

reasoning? In *Proceedings of the 2019 ACL Workshop BlackboxNLP*, 2019.

[45] Michihiro Yasunaga, Hongyu Ren, Antoine Bosselut, Percy Liang, and Jure Leskovec. QA-GNN: Reasoning with language models and knowledge graphs for question answering. In *Proceedings of NAACL-HLT*, 2021.

[46] Tatu Ylonen. Wiktextract: Wiktionary as machine-readable structured data. In *Proceedings of the 13th Language Resources and Evaluation Conference (LREC)*, 2022.

[47] Xikun Zhang, Antoine Bosselut, Michihiro Yasunaga, Hongyu Ren, Percy Liang, Christopher D. Manning, and Jure Leskovec. GreaseLM: Graph REASoning enhanced language models. In *International Conference on Learning Representations (ICLR)*, 2022.

[48] Zhengyan Zhang, Xu Han, Zhiyuan Liu, Xin Jiang, Maosong Sun, and Qun Liu. ERNIE: Enhanced language representation with informative entities. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019.

A Cross-lingual models on their training language

(a) XLM-R-base trained on SICK, English tests

	Accuracy					MCC				
System	SICK	AX-b	Break.	Conj.	FraCaS	SICK	AX-b	Break.	Conj.	FraCaS
[CLS]	90.2 ±1.0	57.2 ±1.0	67.2 ±8.1	44.1 ±2.0	61.8 ±1.3	76.0 ±2.3	18.0 ±1.7	29.4 ±7.3	10.1 ±3.0	13.3 ±3.1
K-BERT										
KGAnet										
KGNLI										
COVER	**90.7** ±0.3	**60.6**† ±2.3	**74.8** ±6.3	**55.3**† ±3.4	**62.2** ±1.8	**77.5** ±0.7	**23.3**† ±3.0	**42.6**† ±5.8	**21.8**† ±3.4	**15.5**† ±3.5

(b) XLM-R-base trained on InferBR, Portuguese tests

	Accuracy				MCC			
System	InferBR	ASSIN	ASSIN 2	SICK-BR	InferBR	ASSIN	ASSIN 2	SICK-BR
[CLS]								
K-BERT								
KGAnet								
KGNLI								
COVER								

Table 6: Cross-lingual models of Table 2 (c, d) evaluated on the test sets of their training language (same format as Table 2: accuracy and MCC of the binary decision, ×100, mean over 10 seeds). †Paired Wilcoxon $p < 0.05$ against [CLS] on the same metric.